mplementation and verification of 7-segment

bjectives: To understand the working of a 7 segment and display its coding mechanism.

eory:

ost often seven segment displays are used to display the digits in gital watches, calculators, clocks, measuring instrument and digital counter ..

There are two types of LED 7-segment display called.

ommon cathode

ommon anode

CD to Seven Segment decoder

CD to seven segment decoder has four inputs (A, B, C, D) and 7 output es (a, b, c, d, e, f, g) this output is given to seven segment LED display.

th table:

tal	Input lines				Output lines							Display
	A	B	C	D	a	b	c	d	e	f	g	Pattern.
	0	0	0	0	1	1	1	1	1	1	0	
	0	0	1	0	0	1	1	0	0	0	0	
	0	0	1	0	1	1	0	1	1	0	1	
	0	0	1	1	1	1	1	1	0	0	1	
	0	1	0	0	0	1	1	0	0	1	1	
	0	1	0	1	1	0	1	1	0	1	1	
	0	1	1	0	1	0	1	1	1	1	1	
	0	1	1	1	1	1	1	0	0	0	0	
	1	0	0	0	1	1	1	1	1	1	1	
	1	0	0	1	1	1	1	1	0	1	1	

K-maps

for a

AB \ CD	00	01	11	10
00	1	0	1	1
01	0	1	1	1
11	X	X	X	X
10	1	1	X	X

$= A + BC + BD + B'D'$

for b

AB \ CD	00	01	11	10
00	1	1	1	1
01	1	0	1	0
11	X	X	X	X
10	1	1	X	X

$= B' + C'D' + CD$

for c,

AB \ CD	00	01	11	10
00	1	1	1	0
01	1	1	1	1
11	X	X	X	X
10	1	1	X	X

$= B + DC'$

for d,

AB \ CD	00	01	11	10
00	1	0	1	1
01	0	1	0	1
11	X	X	X	X
10	1	1	X	X

$= A + \bar{B}D + BC + C\bar{D} + BCD$

for e,

AB \ CD	00	01	11	10
00		1	0	1
01		0	0	1
11		X	X	X
10		1	0	X

$= C\bar{D} + \bar{B}\bar{D}$

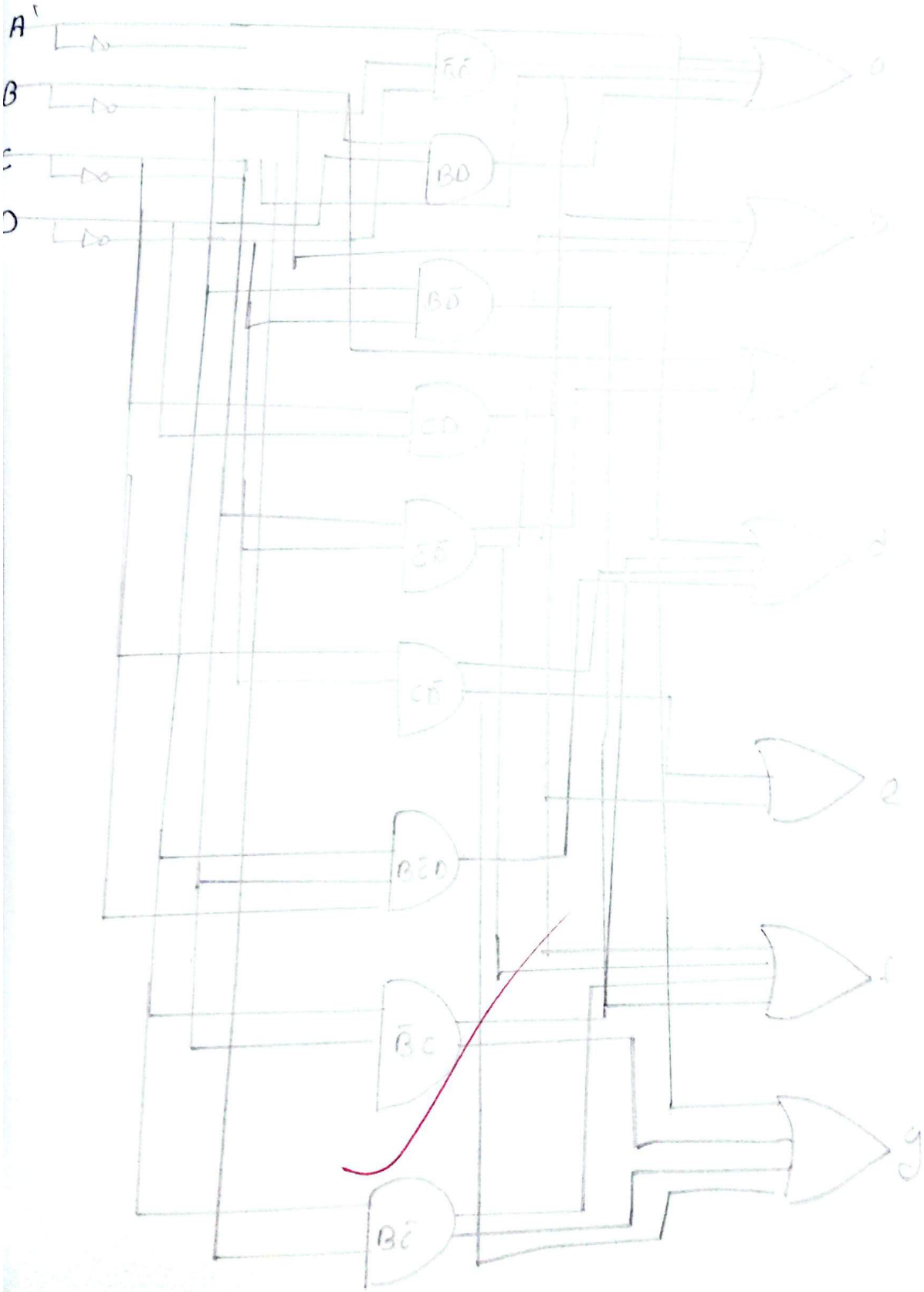
for f,

AB \ CD	00	01	11	10
00	1	0	0	0
01	1	1	0	1
11	X	X	X	X
10	1	1	X	X

$= A + B\bar{C} + B\bar{D} + \bar{C}\bar{D}$

for g,

AB \ CD	00	01	11	10
00	0	0	1	1
01	1	1	0	1
11		X	X	X
10		1	X	X

$= A + \bar{B}C + B\bar{C} + C\bar{D}$

Schematic of BCD to 7segment